Bridging the Gap: Statistical vs Machine Learning Methods

Performance, Cost, and Interpretability Trade-offs in the Age of Transparent AI

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Overview

Problem Statement

The Core Question

How do Machine Learning and Statistical models compare across different problem domains in terms of:

- **Performance** (accuracy, R^2, RMSE)
- **Cost** (training time, computational resources)
- **Interpretability** (explainability, statistical inference)

Research Objectives

1. Compare ML and Statistical approaches across multiple problem domains
2. Quantify trade-offs between performance, cost, and interpretability
3. Provide evidence-based recommendations for model selection
4. Connect findings to Interpretable ML research and Transparent AI

Scope of the Study

Category	Count	Details
Problem Domains	4	Regression, Classification, Time Series, Factorial Analysis
Datasets	13	150 to 35,000 samples
ML Models	5	Linear Regression, Random Forest, SVM, RNN/LSTM, Gradient Boosting
Statistical Models	5	OLS Regression, Logistic Regression, ANOVA, ARIMA, Cox/Poisson
Total Comparisons	13	Model-Dataset pairs

Problem Domains & Datasets

- **Regression:** California Housing, Diabetes, Wine Quality, Energy Consumption
- **Classification:** Iris, Wine Classification, Breast Cancer, Credit Default
- **Time Series:** Energy Consumption, Stock Prices, Temperature, Sales Forecasting
- **Factorial Analysis:** Treatment Effects, Educational Outcomes, Agricultural Experiment, Drug Efficacy Study

Regression: Models & Datasets

Statistical Model:

- OLS Regression (statsmodels)

$$Y = X\beta + \varepsilon$$
$$\hat{\beta} = (X^T X)^{-1} X^T Y$$

- Provides p-values, confidence intervals
- Statistical inference capabilities

ML Model:

- Linear Regression (scikit-learn)

$$\hat{y} = f_\theta(X)$$
$$\theta^* = \arg\min_\theta \mathcal{L}(Y, f_\theta(X))$$

- Optimized for prediction
- Fast training and inference

Datasets

- **California Housing:** 20,640 samples, 8 features
- **Diabetes:** 442 samples, 10 features
- **Wine Quality:** 1,599 samples, 11 features
- **Energy Consumption:** 1,000 samples (synthetic), 6 features

Regression: EDA & Preprocessing

Exploratory Data Analysis

- Distribution analysis (histograms, box plots)
- Correlation matrices
- Missing value detection
- Outlier identification
- Feature-target relationships

Preprocessing Steps

- Train-test split (80-20)
- Standardization: $z = \frac{x-\mu}{\sigma}$
- Feature scaling for numerical stability
- Handling of skewed distributions
- Outlier treatment (if needed)

Regression: Results

Dataset	OLS R^2	ML LR R^2	OLS Time	ML Time
California Housing	0.575	0.575	0.0001s	0.0005s
Diabetes	0.453	0.453	0.0001s	0.0005s
Wine Quality	0.787	0.787	0.0001s	0.0005s
Energy Consumption	0.811	0.970	0.0001s	0.0005s

OLS Advantages:

- Statistical inference (p-values)
- Confidence intervals
- Model diagnostics (F-test, AIC, BIC)
- 5x faster training

ML Advantages:

- Similar performance on linear data
- Better on complex patterns (Energy)
- Optimized for prediction speed
- Easier to integrate in pipelines

Factorial Analysis: Models & Datasets

Statistical Model:

- ANOVA (Analysis of Variance)

$$Y_{ij} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \varepsilon_{ij}$$

$$F = \frac{\mathrm{MS}_{\text{between}}}{\mathrm{MS}_{\text{within}}}$$

- Tests null hypothesis: $H_0 : \mu_1 = \mu_2 = \ldots = \mu_k$
- Provides F-statistics and p-values

ML Model:

- Support Vector Regression (SVM)

$$\min_{\mathrm{w},b} \quad \frac{1}{2}\|\mathrm{w}\|^2 + C\sum_{i=1}^{n} \xi_i$$

$$K(\mathrm{x}_i, \mathrm{x}_j) = \phi(\mathrm{x}_i)^T \phi(\mathrm{x}_j)$$

- RBF kernel for non-linear relationships
- Hyperparameter tuning via grid search

Datasets

- **Treatment Effects:** Clinical trial data
- **Educational Outcomes:** Teaching method comparison
- **Agricultural Experiment:** Fertilizer $\times$ Water level (3$\times$3 factorial)
- **Drug Efficacy Study:** Drug treatment comparison

Factorial Analysis: EDA & Preprocessing

Exploratory Data Analysis

- Group means and standard deviations
- Box plots by treatment groups
- Interaction plots (for factorial designs)
- Assumption checking (normality, equal variances)
- Residual analysis

Preprocessing

- Categorical encoding for group factors
- Feature engineering (interaction terms for ANOVA)
- Standardization for SVM
- Train-test split (80-20)
- Hyperparameter tuning for SVM (Grid Search CV)

Factorial Analysis: Results

Dataset	ANOVA R^2	SVM R^2	ANOVA Time	SVM Time
Treatment Effects	0.417	0.513	0.0005s	2.91s
Educational Outcomes	-0.002	0.284	0.001s	0.19s
Agricultural Experiment	0.464	0.745	0.001s	0.23s
Drug Efficacy Study	0.728	0.880	0.001s	0.25s

ANOVA Advantages:

- Interpretable F-statistics
- P-values for hypothesis testing
- Handles factorial designs
- Very fast (100-1000x faster)

SVM Advantages:

- Higher prediction accuracy
- Captures non-linear patterns
- Better on complex interactions
- More flexible feature use

Classification: Models & Datasets

Statistical Model:

- Logistic Regression (statsmodels)

$$P(Y=1|X) = \frac{1}{1+e^{-X\beta}}$$

$$\log\left(\frac{p}{1-p}\right) = X\beta$$

- Maximum Likelihood Estimation
- Provides odds ratios, p-values

ML Model:

- Random Forest (scikit-learn)

$$\hat{y} = \frac{1}{B}\sum_{b=1}^{B} T_b(X;\theta_b)$$

$$T_b \sim \text{Decision Tree}$$

- Ensemble of B trees
- Feature importance scores

Datasets

- **Iris:** 150 samples, 3 classes, 4 features
- **Wine Classification:** 178 samples, 3 classes, 13 features
- **Breast Cancer:** 569 samples, 2 classes, 30 features
- **Credit Default:** 1,000 samples (synthetic), 2 classes, 10 features

Classification: EDA & Preprocessing

Exploratory Data Analysis

- Class distribution analysis
- Feature distributions by class
- Correlation analysis
- Pair plots for feature relationships
- Class imbalance detection

Preprocessing

- Train-test split (80-20)
- Standardization (for Logistic Regression)
- Handling class imbalance (if needed)
- Feature selection (optional)
- Hyperparameter tuning for Random Forest

Classification: Results

Dataset	Logistic Acc.	RF Acc.	Logistic Time	RF Time
Iris	1.000	1.000	0.002s	0.15s
Wine Classification	0.956	0.982	0.002s	0.15s
Breast Cancer	0.956	0.965	0.002s	0.15s
Credit Default	0.999	0.999	0.002s	0.15s

Logistic Regression:

- Excellent on linearly separable data
- Interpretable coefficients
- P-values and confidence intervals
- 75x faster training

Random Forest:

- Better on complex patterns
- Handles non-linear relationships
- Feature importance scores
- More robust to outliers

Time Series: Models & Datasets

Statistical Model:

- ARIMA (AutoRegressive Integrated Moving Average)

$$\phi(B)(1-B)^d X_t = \theta(B)\varepsilon_t$$
$$\text{ARIMA}(p, d, q)$$

- Auto-selection of (p, d, q)
- Interpretable coefficients
- Statistical diagnostics

Deep Learning Model:

- RNN/LSTM (PyTorch)

$$h_t = \tanh(W_h h_{t-1} + W_x x_t + b)$$
$$\hat{y}_t = f(h_t)$$

- Captures long-term dependencies
- Handles complex temporal patterns
- Requires more data and computation

Datasets

- **Energy Consumption:** 1,000 time steps (synthetic)
- **Temperature:** 1,000 time steps (synthetic)
- **Sales Forecasting:** 1,000 time steps (synthetic)

Time Series: EDA

Exploratory Data Analysis

- Time series plots
- Autocorrelation Function (ACF)
- Partial Autocorrelation Function (PACF)
- Stationarity tests (ADF test)
- Trend and seasonality detection
- Decomposition (trend, seasonal, residual)

Time Series:Preprocessing

Preprocessing

- Train-test split (temporal)
- Differencing (for ARIMA stationarity)
- Sequence creation (for LSTM)
- Normalization/Standardization
- Lag feature engineering

Evaluation

R^2, RMSE, MAE on test set forecasts

Time Series: Results

Dataset	ARIMA R^2	RNN/LSTM R^2	ARIMA Time	LSTM Time
Energy Consumption	0.450	0.890	0.18s	2.5s
Temperature	0.972	0.965	0.18s	2.5s
Sales Forecasting	0.972	0.969	0.18s	2.5s

ARIMA Advantages:

- Interpretable coefficients
- Statistical diagnostics
- 14x faster training
- Excellent on stationary data

LSTM Advantages:

- Better on complex patterns
- Handles non-stationary data
- Captures long dependencies
- Superior on Energy dataset

Overall Performance Summary

Metric	Statistical	ML/DL
Avg Performance (R^2/Acc)	0.547	0.818
Avg Training Time	0.037s	0.763s
Efficiency (Performance/Cost)	High	Medium
Interpretability	88.3/100	37.4/100

Statistical Models:

- OLS Regression: 0.650 R^2, 0.0001s
- Logistic Regression: 0.880 Acc, 0.002s
- ANOVA: 0.550 R^2, 0.0001s
- ARIMA: 0.450 R^2, 0.18s

ML/DL Models:

- Random Forest: 0.920 Acc, 0.15s
- SVM: 0.850 R^2, 0.40s
- LSTM: 0.890 R^2, 2.5s
- Gradient Boosting: 0.850 R^2, 0.50s

Performance vs Training Cost Trade-off

CostVsPerformance.png

Performance vs Interpretability Trade-off

The Interpretability-Performance Curve

- Traditional trade-off: Higher performance $\Leftrightarrow$ Lower interpretability
- **Interpretable ML Research:** Moving toward the ideal solution
- Goal: Bridge the gap between statistical and ML approaches

bridging_gap.png

Winners by Category

Category	Winner	Performance
Best Overall Accuracy	Random Forest (ML)	0.920
Fastest Training	OLS Regression (Stat)	0.0001s
Most Interpretable	OLS Regression (Stat)	100/100
Best Efficiency	OLS Regression (Stat)	6435.64
Best for Time Series (Simple)	ARIMA (Stat)	$R^2 = 0.999$
Best for Time Series (Complex)	LSTM (DL)	$R^2 = 0.890$

Key Insight

No single approach dominates all categories. The choice depends on:

- Primary requirement (accuracy vs interpretability vs speed)
- Dataset characteristics (size, complexity, linearity)
- Computational constraints
- Regulatory requirements

The Fundamental Trade-offs

Performance

$$P(\theta) = \mathbb{E}[\mathcal{M}(Y, \hat{Y})] = \frac{1}{n}\sum_{i=1}^{n} \mathcal{M}(y_i, f_\theta(x_i))$$

Metrics: R^2, Accuracy, RMSE

Cost

$$C(\theta) = T_{\text{train}}(\theta) + \lambda M(\theta)$$

Training time + memory cost

Computational complexity

Interpretability

$$I(\theta) = \text{Explainability Score} \in [0, 100]$$

Transparency and trust

Key Insight

The optimization problem:

$$\max_{\theta} \quad P(\theta) - \alpha C(\theta) + \beta I(\theta)$$

s.t. constraints

Efficiency Analysis

Ranking	Model	Efficiency
1	OLS Regression (Stat)	6435.64
2	ANOVA (Stat)	5445.54
3	Linear Regression (ML)	1297.41
4	Logistic Regression (Stat)	439.78
5	Poisson Regression (Stat)	349.65
...	...	...
Last	RNN/LSTM (DL)	0.36

Statistical Models:

- 20-100x more efficient
- Best for real-time applications
- Low computational cost
- Excellent for exploration

ML/DL Models:

- Higher performance potential
- Better for complex patterns
- Suitable for batch processing
- When accuracy is critical

When to Choose Statistical Models

Choose Statistical Models When:

- **Interpretability is critical** - Need p-values, confidence intervals, and statistical inference
- **Regulatory compliance** - Transparent decision-making required (e.g., healthcare, finance)
- **Hypothesis testing** - Understanding causal relationships and testing theories
- **Small to medium datasets** - Optimal performance on limited data
- **Real-time applications** - Fast training and prediction needed
- **Research and development** - Need to understand underlying relationships

Best Models: OLS Regression, Logistic Regression, ANOVA, ARIMA, Cox Regression

When to Choose ML/DL Models

Choose ML/DL Models When:

- **Maximum accuracy is priority** - Need best predictive performance
- **Large, complex datasets** - High-dimensional or non-linear patterns
- **Production systems** - Computational resources available
- **Pattern discovery** - Exploratory analysis on complex data
- **Non-linear relationships** - When linear assumptions fail
- **Sequential/temporal data** - LSTM/RNN for time series with complex patterns

Best Models: Random Forest, SVM, LSTM/RNN, Gradient Boosting

Key Takeaways

1. **No universal winner** - Both approaches have distinct advantages
2. **Statistical models excel in:**
   - Interpretability (88.3/100 vs 37.4/100)
   - Speed (20-100x faster training)
   - Efficiency (6435.64 vs 0.36 efficiency scores)
   - Regulatory compliance and hypothesis testing
3. **ML/DL models excel in:**
   - Complex dataset performance (0.818 vs 0.547 avg)
   - Scalability to large data
   - Capturing non-linear patterns
   - Handling complex temporal dependencies
4. **The optimal choice depends on:**
   - Primary requirement (accuracy vs interpretability vs speed)
   - Dataset characteristics (size, complexity, linearity)
   - Computational constraints
   - Regulatory requirements

Questions & Discussion

Thank You!

Questions?

Contact: ML vs Statistical Models Research Team
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References & Resources

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